

Digital Payments and Financial Inclusion in India: A State-Level Look at the Early and Mature UPI Eras

A State-Level Look at the Early and Mature UPI Eras

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1. Executive summary

Across Indian states in two distinct stretches of UPI's evolution, financial-inclusion programmes that explicitly decouple account access from income leave a measurable footprint on per-capita digital-payment use. Programmes whose access scales with income, bank-office expansion, telecom buildout, urban agglomeration, do not, because their cross-state variation is absorbed by state income. This is the substantive read of two parallel regressions, five years apart, with different dependent variables, that give the same answer.

The two regressions cover the early-UPI era of FY 2019-20 and FY 2020-21, when UPI was one of five rails making up India's digital-payment stack, and the mature-UPI era of April 2023 through January 2026, when UPI alone settled the bulk of retail digital payments. Both use the same four state-level controls (per-capita state domestic product, urban share, PMJDY beneficiaries per adult, bank-office density), the same estimator (pooled OLS with year fixed effects and state-clustered standard errors), and the same log-log specification.

Three findings hold across both eras:

- Per-capita state income is the dominant cross-state correlate of digital-payment use, with an elasticity close to one in both regressions. The two estimates are statistically indistinguishable from each other.
- PMJDY beneficiary density adds explanatory power beyond income, with a positive elasticity statistically distinguishable from zero in both regressions (+0.50 in the early era, +0.34 in the mature era). Conditional on income, states with broader PMJDY enrollment per adult use more digital payments per capita.
- Bank-office density, urban share, and (in the mature era) internet penetration do not survive the income control in either regression. They correlate with digital-payment use unconditionally but not separably from the income gradient.

The cross-state distribution is also narrowing. The coefficient of variation in monthly per-capita UPI declines at roughly 10 percent per year over the 34-month panel; the standard deviation of log per-capita UPI traces a parallel decline. Convergence is real, but a sixfold gap between the highest-use and lowest-use states persists at the end of the panel.

Limits. The findings are descriptive cross-state correlations after controlling for income, not causal effects. The PMJDY result, the firmest conditional finding, is associational. Section 5 sketches the data and identification strategies that would let stronger claims follow.

2. Why this question matters

In January 2026, India processed 21.7 billion UPI transactions worth 28.33 lakh crore. That was a 28 percent increase over the same month a year earlier, and works out to roughly 700 million transactions every

day.¹ UPI is now the largest retail payments system in the world by volume. In a single month it settles more transactions than the card networks of many large economies process in an entire year. The Indian government and the National Payments Corporation of India (NPCI), which operates the rail, regularly point to this scale as evidence that digital payments have meaningfully advanced financial inclusion.

That January 2026 figure is the latest point in a longer story. The story splits into two stretches worth analysing separately. The first is the early-UPI era of FY 2019-20 and FY 2020-21. UPI was one of several rails competing for digital-payment volume in that period. The state-level data publishes only a five-rail composite (BHIM + IMPS + RuPay POS + UPI + USSD). The second is the mature-UPI era from April 2023 through January 2026. By then UPI alone settled the bulk of retail digital payments, and state-level pure-UPI data became consistently available. Across the most recent twelve months (February 2025 through January 2026), the 36 states and union territories of India together processed about 142 billion state-attributable UPI transactions worth 197 lakh crore. That works out to roughly 8.3 transactions per person per month, weighted by population.² This note compares the two eras using a single set of state-level controls. That way, what changed about who uses digital payments, and what stayed the same, is visible in the coefficients.

The national average hides the state-level variation

The national figure of 8.3 monthly transactions per person hides a wide range across states. Tripura sits at the bottom with 3.4. Telangana and Goa sit at the top, both at 22.0. The gap is more than six-fold. The most-used states do about five times the per-capita volume of the most populous low-use states (figure 2, table 1). A national average that hides variation this large cannot tell us whether UPI’s scale has reached people broadly. For policy purposes, the state is the right unit. Payments infrastructure, regulations, and inclusion programs are rolled out and evaluated at the state level. The populations they reach differ in ways a national average smooths over.

Two readings of the gap, both true

The cross-state distribution can be read two ways. A useful policy account of UPI needs to hold both readings at once.

The first reading is that states are converging. Between April 2023 and January 2026, the cross-state coefficient of variation in monthly per-capita UPI fell by roughly ten percent per year. The standard deviation of log per-capita UPI fell at a similar rate (figure 4). In plain terms: low-use states are growing their per-capita UPI faster than high-use states are. The gap is closing.

The second reading is that the gap is still very large. The highest-use states do five to eight times what the lowest-use states do. The gap is narrowing, but it is far from closed.

Why a financial-inclusion lens, specifically

UPI is routinely cited as a financial-inclusion success. But “UPI scaled” and “UPI scaled inclusively” are different claims, and the second is the more demanding one. The inclusion-relevant questions are state-comparable. Do states with more PMJDY beneficiaries, accounts opened under the government’s flagship financial-inclusion scheme, actually use digital payments more? Does the density of bank branches and offices, the older form of financial-rail expansion, predict use in the same way that account access does? Does the income gradient absorb everything else once it is conditioned on? And do the answers look the same in the early and mature UPI eras, or has the relationship shifted as the rail matured? A two-period state-level analysis can answer these questions; a national time series cannot.

¹National Payments Corporation of India, UPI Product Statistics, <https://www.npci.org.in/what-we-do/upi/product-statistics>. The January 2026 figures are reported in NPCI’s monthly press release and in the business press (e.g., DD News and Business Standard, 1 February 2026).

²Author’s calculation from the panel constructed for this note. NPCI’s all-India figure exceeds the state-summed total because a non-trivial share of transactions are reported under “Unclassified” and cannot be attributed to a specific state. The implications for the state-level analysis are discussed in section 3.

What this note does

Section 3 sets the empirical stage with the cross-state snapshot for the most recent twelve months and the trend across the full 34 months of mature-era data. Section 4 presents three findings about what predicts state-level digital-payment use, each supported by two parallel regressions, one for the early-UPI era (FY 2019-20 and FY 2020-21) using the five-rail composite, one for the mature-UPI era (April 2023 through January 2026) using pure UPI data. Both regressions use the same set of state-level controls so the cross-era pattern is visible in the coefficients. Section 5 takes up what the findings imply for digital-payments and financial-inclusion policy. Methodology, full regression tables, and a diagnostic on one variable (literacy) that we set aside on measurement-validity grounds are in Appendix A; data sources and construction notes are in Appendix B.

3. The state-level landscape

This section sets the empirical stage in two parts. The first half is the cross-state snapshot at the most recent twelve months for which we have data, February 2025 through January 2026. The second half is the trend across the full 34 months of mature-era data, asking whether the cross-state distribution has been narrowing, widening, or holding steady. The findings in Section 4 are read against this landscape.

The state ranking

Figure 2 ranks the 36 states and union territories by their average monthly per-capita UPI transactions over the most recent twelve months (February 2025 – January 2026), coloured by income quartile. Telangana sits at the top with 22.0 monthly transactions per person, essentially tied with Goa (22.0). Delhi (21.7) and Chandigarh (21.4) follow closely; Maharashtra is fifth at 16.6. The bottom of the distribution is dominated by the most populous low-income states and the small north-eastern states: Tripura (3.4), Bihar (3.5), Manipur (4.4), Jharkhand (4.4), and Meghalaya (4.4). The India population-weighted average, the figure a national reader would normally cite, is 8.3 transactions per person per month, marked on the figure by the dashed vertical line.

The income-tier colouring makes the cross-state pattern visible at a glance. The upper half of the ranking is overwhelmingly the dark blue of the upper income quartile, and the lower half is overwhelmingly the dark red of the lowest quartile. Two pairs of states sit visibly off the diagonal. Gujarat and Himachal Pradesh, both upper-mid-income states, show up in the lower half of the per-capita UPI ranking, well below where their income tier alone would predict. Andhra Pradesh and Arunachal Pradesh, both in the lower-mid-income tier, show up in the upper half, well above what their income tier alone would predict. These four are visible exceptions to the broad income–UPI association, but they remain exceptions: across the full set, income tier is the dominant predictor of where a state sits, which is what the regression in Section 4 quantifies formally.

The summary statistics

Table 1 summarises the seven variables that enter the cross-state analysis: per-capita UPI use (the dependent variable), the five regression controls (per-capita NSDP, internet density, urban share, PMJDY beneficiaries per adult, bank-office density), and the appendix-only literacy variable. For each variable we report the cross-state mean, standard deviation, minimum and maximum, and the population-weighted India-wide value.

Three patterns in the table are worth pulling out, because they illustrate how cross-state averages can diverge from population-weighted national figures and what that divergence reveals.

Per-capita UPI use has a state mean of 9.6 monthly transactions per person but a population-weighted India figure of 8.3. The gap is small but goes in a specific direction: the highest-use states are mostly not the most populous ones. Telangana, Chandigarh, Delhi, and Goa are smaller in population than their UPI rank would suggest, while Uttar Pradesh and Bihar, large and populous, sit toward the bottom. The state mean overstates the typical Indian’s per-capita UPI use; the population-weighted figure is the more honest reading of the national level.

PMJDY beneficiaries per adult shows the opposite pattern. The state mean is 0.41 accounts per adult, but the population-weighted figure is 0.53. The difference is large and reflects exactly what PMJDY’s targeting was designed to do: enrollment is highest in populous low-income states (Bihar at 0.75, Jharkhand at 0.69, Madhya Pradesh at 0.68, Rajasthan at 0.61, Uttar Pradesh at 0.59). When we weight by population, those states pull the national figure up well above the cross-state mean. The same scheme that looks modest as a state-level mean (41 percent of adults) is a much larger national effort when measured in adult-weighted terms (53 percent of adults).

Bank-office density per 100,000 population shows the same direction as per-capita UPI use: state mean of 17.9 per 100k, population-weighted India figure of 12.4. Small states and union territories have many more bank offices per person than large states do (Goa at 46, Lakshadweep at 40, Chandigarh at 39, Ladakh at 35, Sikkim at 27, Himachal Pradesh at 25; the largest states sit between 8 and 12). The state-mean figure should not be mistaken for the average Indian’s access to a bank office, it overstates that average by close to 50 percent.

The remaining variables move less between state mean and population-weighted figure because they are more uniformly distributed: urban share (46.5 percent state mean, 36.1 percent population-weighted; the population-weighted figure pulls down because rural states are populous), literacy (85.6 vs 80.8 percent), internet density (80.4 vs 68.6 subscribers per 100), and per-capita NSDP (146,000 vs 116,000).

And the gap is closing

The snapshot is at the most recent twelve months, but the cross-state distribution has been moving over the full 34-month window for which we have monthly state-level UPI data, April 2023 through January 2026. Figure 4 plots two complementary dispersion measures over that window, each indexed to its own value at panel start. Both narrow steadily.

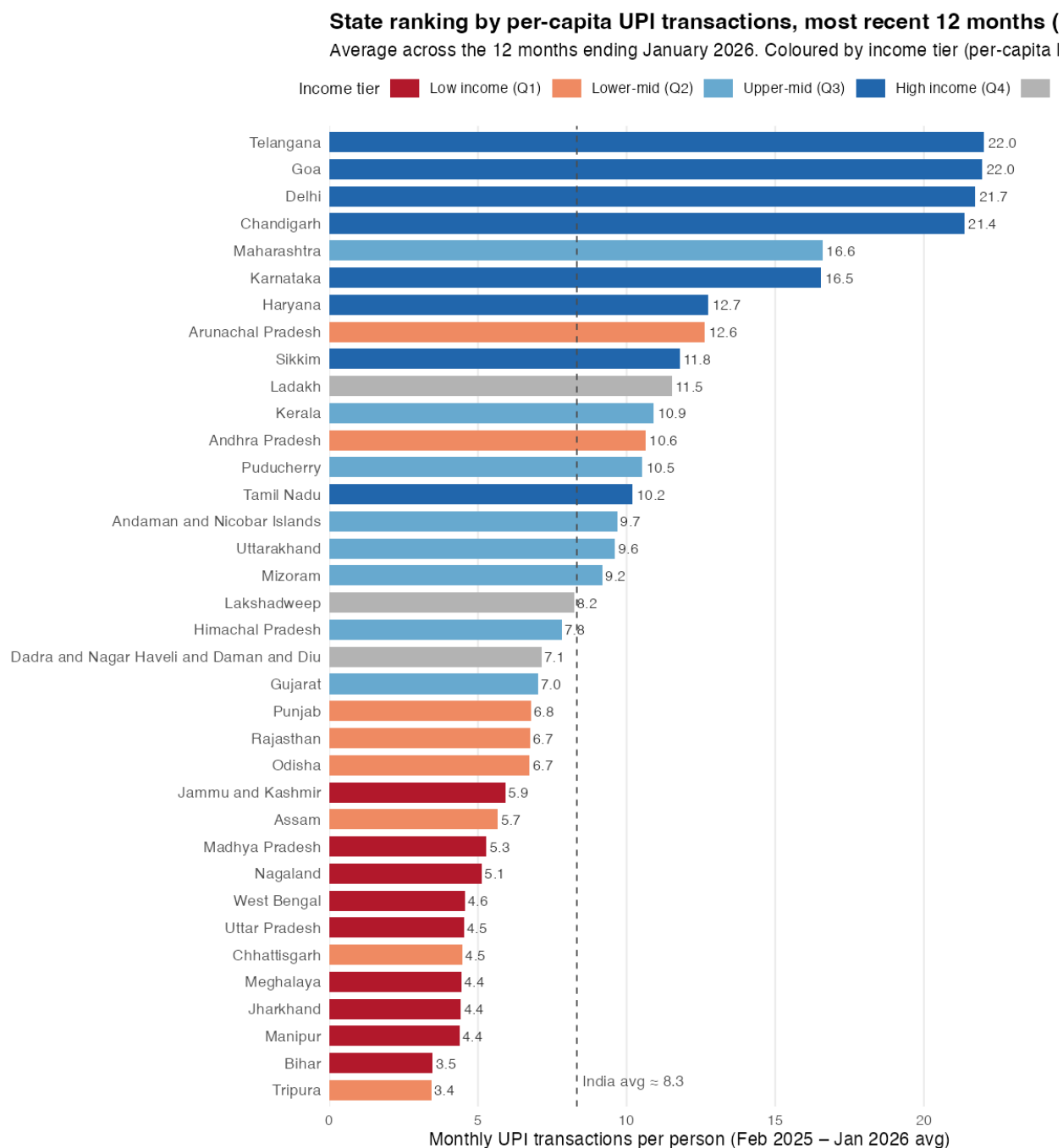
The cross-state coefficient of variation in monthly per-capita UPI declines at roughly 10 percent per year, ending the panel about 27 percent below its April 2023 starting value. The standard deviation of log per-capita UPI declines at roughly 8 to 9 percent per year, ending about 24 percent below its starting value. The two measures track each other closely, the convergence finding does not depend on which one you choose. In plain terms: low-use states are growing their per-capita UPI faster than high-use states are. The gap is closing.

A useful way to think about the magnitude. At the panel-start level of dispersion, the highest-use state did about ten times the per-capita volume of the lowest-use state. By January 2026 that ratio had fallen closer to seven. The gap is still large, convergence is not parity, but the trend over thirty-four months is steady downward narrowing. Two robustness checks on the convergence pattern are in Appendix B: a test of whether the growing “Unclassified” share of national UPI volume is concentrated in particular states (it is not, statistically), and a note on Maharashtra’s six-percentage-point share loss over the panel period (which contributes to the narrowing but does not drive it).

Reading these levels into the findings

The level of cross-state variation in per-capita UPI use is large. A more-than-six-fold gap between the lowest- and highest-use states is not a rounding error around the national average. The convergence trend is real, but it is operating against an initial dispersion of meaningful magnitude. Section 4 takes up which state-level variables predict where states sit in this distribution.

Figure 2 — State ranking by per-capita UPI use, most recent twelve months (Feb 2025 – Jan 2026)

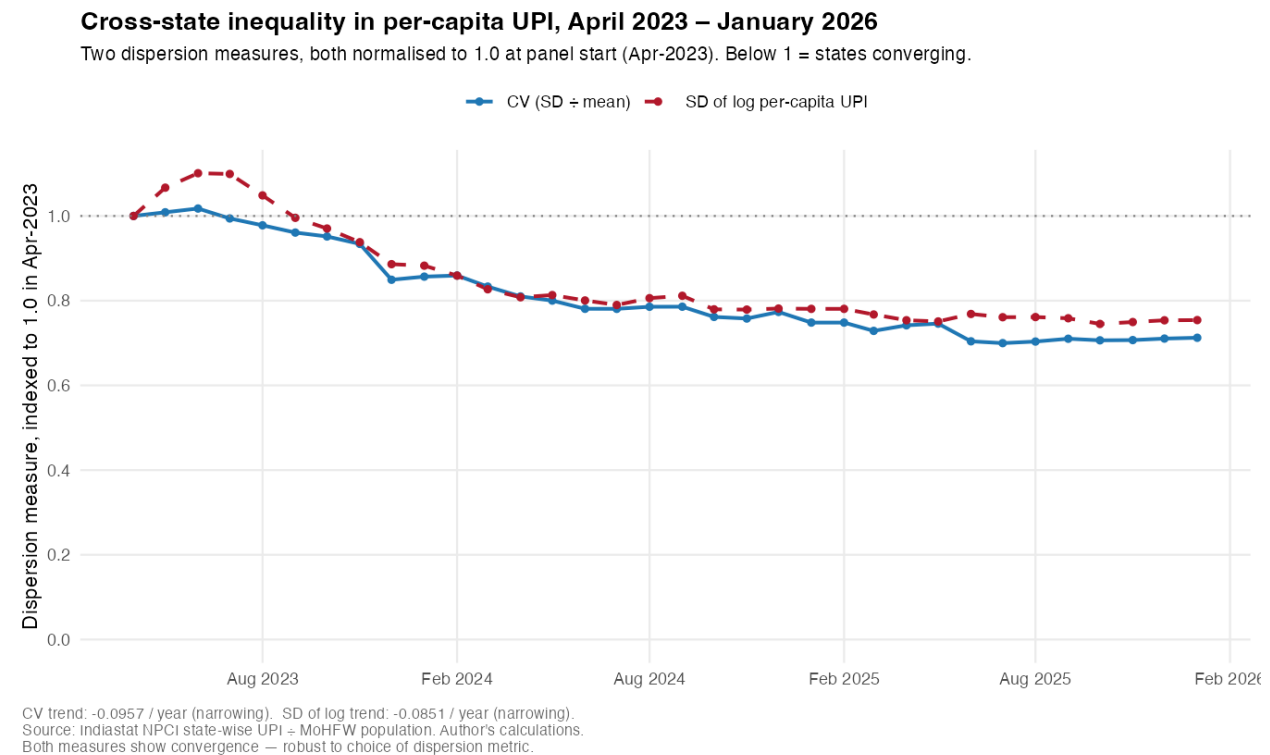


Source: Indiatat NPCI state-wise UPI (rolling 12-month window ending Jan 2026); MoHFW population mid-period (mean of 1-Jul-2025, 1-Jul-2026); per-capita NSDP at constant 2011-1
Author's calculations.

Table 1 — Descriptive statistics, most recent twelve months (Feb 2025 – Jan 2026)

Variable	Unit	N	Mean	SD	Min	Max	India (pop-wt.)
Per-capita UPI transactions (Feb 2025 – Jan 2026 avg)	per person/month	36	9.6	5.5	3.4	22.0	8.3
Per-capita NSDP (latest available, constant 2011-12 prices)	INR	33	146,389	76,268	36,342	357,611	116,350
Internet subscribers (31-Mar-2025, latest TRAI release)	per 100 persons	36	80.4	27.9	41.6	161.8	68.6
Urban population share (interpolated to 2025)	%	35	46.5	25.1	10.4	100.0	36.1
PMJDY beneficiaries per adult (snapshot 13-Dec-2025)	accounts / adult	36	0.41	0.20	0.09	0.91	0.53
Bank offices per 100,000 population (31-Dec-2025)	per 100k pop	36	17.9	9.6	6.5	46.2	12.4
Literacy rate, persons 7+ (FY 2023-24, appendix only)	%	36	85.6	7.1	72.6	98.2	80.8

Figure 4 — Cross-state dispersion in per-capita UPI, Apr 2023 – Jan 2026



4. What predicts state-level digital-payment use

This section presents three findings about what predicts state-level digital-payment use across Indian states. Each finding is supported by two parallel regressions: the early-UPI era of FY 2019-20 and FY 2020-21, when UPI was one of five rails making up India's digital-payment stack, and the mature-UPI era of April 2023 through January 2026, when UPI alone settled the bulk of retail digital payments. Both regressions use the same four state-level controls (per-capita state domestic product, urban population share, PMJDY beneficiaries per adult, and bank-office density), the same estimator (pooled OLS with year fixed effects and

standard errors clustered at the state level), and the same log–log specification. The mature-era regression adds a fifth control, internet density, that becomes available only in that period.

Full methodology, regression specifications, and detailed result tables are in Appendix A. Table 2 below summarises the headline coefficients on the four shared controls in both eras; the rest of this section walks through what the table makes visible.

Table 2, Cross-era coefficients on the four shared controls

Control	Early-UPI era (FY 2019-20 + 2020-21)	Mature-UPI era (Apr 2023 – Jan 2026)
log(per-capita NSDP)	+1.17 **	+1.05 ***
log(urban share)	+0.25	+0.06
log(PMJDY beneficiaries / adult)	+0.50 *	+0.34 **
log(bank offices / 100k pop)	−0.17	−0.19
Sample size (state-period observations)	66	99

Bold figures are statistically distinguishable from zero. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Coefficients are elasticities, a 1 percent change in the control is associated with a -percent change in per-capita digital-payment use, holding the other controls constant.

Finding 1: Income is the dominant correlate, in both eras

The cleanest empirical pattern in the two regressions is the income elasticity. In the early-UPI era it is +1.17, statistically significant at the 5 percent level; in the mature-UPI era it is +1.05, significant at the 1 percent level. Both are close to a unit elasticity. A 10 percent increase in a state’s per-capita state domestic product is associated with about a 10 percent increase in its per-capita digital-payment use, and that relationship was already in place in 2019-21 and persisted essentially unchanged through 2024-25.

The two estimates are close enough that they are not statistically distinguishable from each other. This is meaningful in itself. The dependent variable in the early era was a five-rail composite (BHIM, IMPS, RuPay-on-POS, UPI, USSD); in the mature era it is pure UPI. The sample sizes differ; the panel structures differ; the controls differ in one variable. Yet the income-elasticity is the same. Whatever else changed about UPI between 2019 and 2026, the sheer scale of transactions, UPI’s displacement of the four other rails, the doubling of bank-office counts, the addition of hundreds of millions of PMJDY beneficiaries, the income gradient of digital-payment use has held its shape.

Two readings follow. First, income is by far the most reliable predictor of where a state will sit in the cross-state distribution of digital-payment use, and any policy aimed at expanding digital payments at the state level is operating against a strong income gradient. Second, the durability of the relationship across two distinct measurement regimes makes it unlikely that the income gradient is an artefact of how the dependent variable is constructed in either era, it is a feature of the underlying state-level distribution. The pattern is associational, not causal; what it does establish is that any other state-level variable claimed to matter for digital-payment use needs to survive controlling for income. Findings 2 and 3 are both about which variables clear that bar.

Finding 2: Financial-inclusion programs (PMJDY) add explanatory power beyond income, in both eras

This is the policy-relevant finding of the analysis. The coefficient on log(PMJDY beneficiaries per adult) is positive and statistically distinguishable from zero in both regressions: +0.50 in the early-UPI era (significant at the 10 percent level) and +0.34 in the mature-UPI era (significant at the 5 percent level). A 10 percent

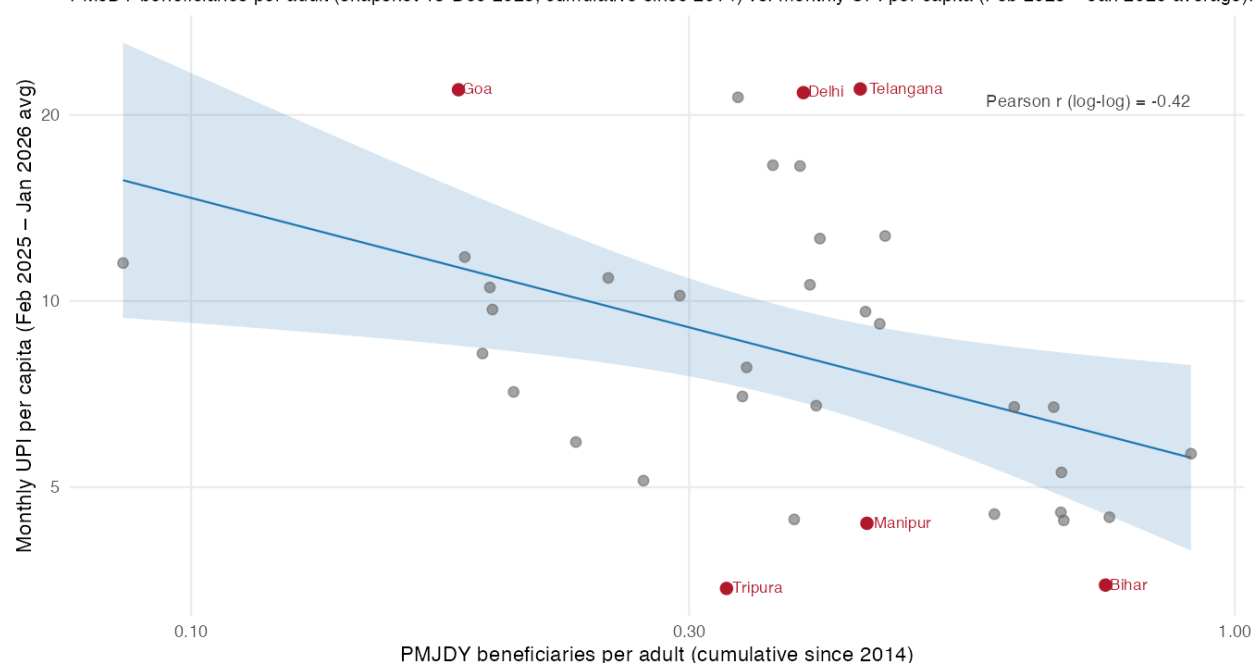
relative increase in PMJDY enrollment per adult is associated with a 3 to 5 percent increase in per-capita digital-payment use, holding income, urban share, bank-office density, and (in the mature era) internet density constant. The two estimates are statistically indistinguishable from each other.

The finding has more bite than the magnitudes alone suggest, because the univariate correlation between PMJDY enrollment and per-capita digital-payment use is negative across Indian states (Pearson r in log-log of about -0.42 ; see Figure 5). PMJDY-heavy states are poorer states, and poorer states use digital payments less. So unconditionally, more PMJDY enrollment looks associated with less digital-payment use. The negative univariate correlation is income masquerading as PMJDY.

Once we control for income, the sign flips. Among states at similar income levels, those with higher PMJDY enrollment per adult use more digital payments. The flip is not a regression artefact, it is what the data say once the income gradient is held constant. And the flip occurs in both eras: when PMJDY had been running for five to six years and was still in its build-out phase (early-UPI era), and when it had reached close to a decade of cumulative enrollment (mature-UPI era).

PMJDY targeting reach vs. digital payments use, by state

PMJDY beneficiaries per adult (snapshot 13-Dec-2025, cumulative since 2014) vs. monthly UPI per capita (Feb 2025 – Jan 2026 average).



Source: PMJDY portal (<https://pmjdy.gov.in/statewise-statistics>) via Wayback Machine snapshot 13-Dec-2025; Indiatat NPCI state-wise UPI (Feb 2025 – Jan 2026 monthly); MoHFW population mid-period; Adult share derived from PLFS 2023-24 Table 1 sample composition (Jul 2023 – Jun 2024). Author's calculations. PMJDY beneficiary totals are cumulative since the scheme's 2014 launch. Correlation is descriptive, not causal.

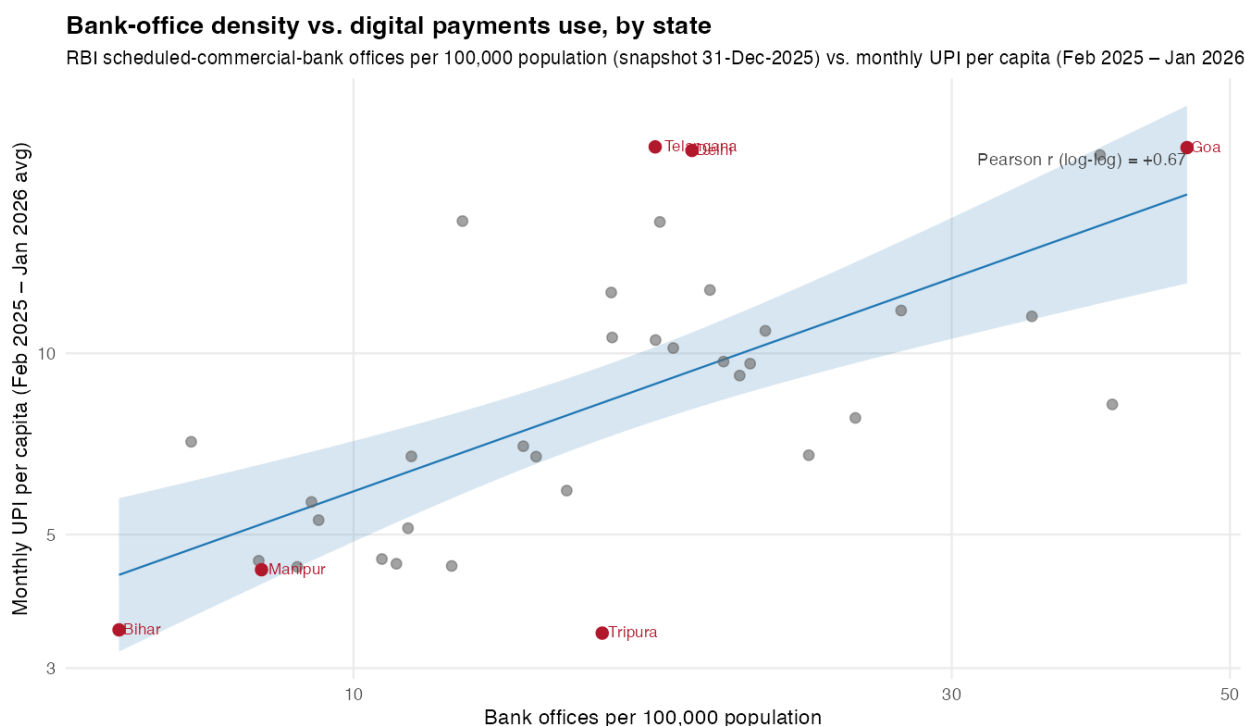
The mechanism is account access. PMJDY's design opens a basic savings account for unbanked households, and an account is the necessary condition for using a digital-payment rail at all. The variable measures the cumulative reach of that scheme, normalised by adult population. The regression detects that this measure of account access predicts digital-payment use beyond what income alone does, in both eras. The reading is not that PMJDY caused UPI adoption, the data here cannot establish that, but that programmatic financial inclusion has a durable association with digital-payments use that survives the income control.

The policy implication, developed in Section 5, is straightforward: financial-inclusion programs that decouple account access from income, as PMJDY does, by deliberately targeting the unbanked, leave a statistical footprint in cross-state digital-payment use. The same is not true of every variable that captures financial inclusion, as Finding 3 shows.

Finding 3: Financial-inclusion infrastructure (bank offices) does not add beyond income, in either era

The contrast with PMJDY is sharp. The coefficient on $\log(\text{bank offices per } 100,000 \text{ population})$ is small and statistically indistinguishable from zero in both regressions: -0.17 in the early era, -0.19 in the mature era. Bank-office density does not survive joint inclusion with income. The same is true of urban share ($+0.25$ and $+0.06$, neither significant) and, in the mature era where it is available, internet density ($+0.34$, also not significant).

This is not because bank-office density is irrelevant to digital-payment use. The univariate correlation between bank-office density and per-capita UPI use is positive and substantial (Pearson r in log-log of about $+0.67$; Figure 6). States with more banking infrastructure per person do use more digital payments per person, in the unconditional cross-section.



Source: RBI Database on Indian Economy, Other STRBI Table 13 (State-wise Number of Functioning Offices of Commercial Banks, 31-Dec-2025); Indiatat NPCI state-wise UPI (Feb 2025 – Jan 2026 monthly); MoHFW population mid-period. Author's calculations. Correlation is descriptive, not causal.

The reason the conditional coefficient is null is collinearity with state income. In the cross-section of Indian states, bank-office density correlates with per-capita NSDP at roughly $+0.85$; urban share correlates with NSDP at roughly $+0.7$; internet density correlates with NSDP at $+0.86$. All three variables capture economically real and theoretically distinct channels, physical banking infrastructure, urban agglomeration, and telecom buildout, but they all move tightly with state income. Once income is in the regression, the marginal contribution of any of these variables beyond what NSDP already absorbs is statistically indistinguishable from zero.

PMJDY survives this test for one structural reason. PMJDY enrollment per adult correlates with per-capita NSDP at roughly -0.62 , strongly negatively. The scheme deliberately targets the unbanked, who are concentrated in poorer states, so its beneficiary density runs in the opposite direction from income. That negative correlation is precisely what gives PMJDY identification beyond income: conditioning on income does not collapse PMJDY's variation, because income and PMJDY move in opposite directions across states.

The substantive distinction this draws is between two kinds of financial inclusion. Financial-inclusion infrastructure, bank offices, telecom buildout, urban agglomeration, is allocated by markets and grows with state

income. In our data, it shows up through the income gradient, not beyond it. Financial-inclusion programs designed to reach the population outside that gradient, PMJDY in our case, show up beyond income in a way the regression can identify separately. This is the policy-relevant distinction the next section takes up.

A note on what we set aside: a sixth state-level variable, literacy, was tried in both eras and produced an interpretable pattern that we read as a measurement-validity problem with the literacy variable as conventionally defined rather than a substantive finding about literacy and digital payments. The full diagnostic is in Appendix A.

5. Policy implications

The findings in Section 4 carry three specific implications for digital-payments and financial-inclusion policy. We take each in turn.

1. Design financial-inclusion programmes to decouple access from income

The actionable read of the PMJDY result is that the kind of inclusion design matters, not just the inclusion variable itself. Programmes that deliberately decouple access from income, as PMJDY does, by targeting the unbanked, leave a measurable footprint on cross-state digital-payment use beyond what general economic development would already produce. Programmes whose access scales with the market, bank-office expansion, telecom buildout, urban agglomeration, do not, because their cross-state variation is absorbed by the income gradient. The mechanism is developed in Section 4, Finding 3.

The general claim that follows is not a finding about one scheme. It is empirical evidence that the design of an inclusion programme, whether it allocates by market or by deliberate counter-targeting, predicts whether the programme will leave a measurable cross-state footprint on digital-payment use. Programmes whose allocation rule moves with state income will, in cross-state monitoring, look like income through a different lens. Programmes that decouple from income may show up separately.

The methodology generalises and the variable does not. Future work should run the same cross-era, same-controls regressions on other state-level inclusion programmes whose enrollment data exist: Direct Benefit Transfer (DBT) volumes, Mudra credit coverage, Aadhaar-enabled Payments System (AePS) uptake, and the targeted merchant-onboarding programmes that BHIM and the National Payments Corporation of India have run since 2017. Each of these has a state-level enrollment count that could be measured against per-capita digital-payment use conditional on income. The methodology in Appendix A is designed to be transportable to any of them.

2. State-level dispersion belongs in inclusion monitoring

The monthly all-India UPI volume that NPCI publishes, and that the press and the government routinely cite, is the wrong figure for inclusion monitoring. It averages over precisely the cross-state variation that is the inclusion-relevant signal. A 21.7-billion-transaction national headline (January 2026) coexists with an eight-fold gap between the highest- and lowest-use Indian states. The headline tells you about UPI's overall scale; it tells you essentially nothing about whether that scale has reached the populations who were the target of inclusion policy in the first place.

The cross-state coefficient of variation and the standard deviation of log per-capita UPI are policy-relevant metrics. Both are easy to compute monthly from the data NPCI already publishes, and both convey information the national headline does not (Section 3, Figure 4). NPCI and the Reserve Bank of India should publish these (or equivalent) dispersion measures alongside the national volume in the monthly UPI release. The inclusion-policy ecosystem cannot monitor distributional progress on a metric that has not been made standard.

A linked data-quality concern: the share of monthly UPI volume reported under “Unclassified”, transactions that cannot be attributed to a specific state, has grown from approximately 12 percent in April 2023 to nearly 40 percent by January 2026. If two-fifths of national UPI volume sits outside the state-level data, the state-level inclusion picture is partially blind. Whatever the cause, Mumbai-routed payment infrastructure,

multi-state merchant attribution, or something else, NPCI’s reporting choices on the Unclassified bucket are an inclusion-policy concern, not just a data engineering one.

3. What this paper does not license, and what would strengthen the claim

The findings here are cross-state and descriptive. They cannot establish causation. Specifically: the bank-office null is a statement about cross-state identification, not evidence that an additional bank office in any particular place would not raise digital-payment use; the same applies to the internet null. The PMJDY result, the firmest conditional finding, is associational, we cannot rule out that PMJDY enrollment and per-capita digital-payment use are jointly driven by state-level political, cultural, or historical factors not in the regression.

Three extensions would substantially strengthen what we can claim. District-level digital-payment data, not currently published at the granularity we have for state level, would allow within-state analysis that holds those unobservable factors constant. Causal designs that exploit the state-by-state phasing of PMJDY’s rollout between 2014 and 2020 could turn the associational finding into an identified treatment effect. A state-level measure of digital literacy, smartphone navigation, app fluency, comfort with numeric prompts, distinct from the basic primary-language literacy variable our data contain, would resolve the puzzle documented in Appendix A and possibly identify a separate channel the current analysis cannot.

Until those data exist, the paper provides robust descriptive evidence on cross-state correlates of digital-payment use across two distinct UPI eras. The policy reads in this section follow from those patterns; they do not exceed them.

6. Conclusion

The question this paper opened with was specific: across Indian states, in two distinct stretches of UPI’s evolution, what predicts state-level digital-payment use, and what does the cross-era pattern imply for inclusion policy?

The answer is built around three findings that hold in both eras. Income is the dominant cross-state correlate of digital-payment use, with a roughly unit elasticity in both regressions. PMJDY beneficiary density adds explanatory power beyond income, with a positive elasticity that is statistically distinguishable from zero in both eras and quantitatively similar across them. Bank-office density, urban share, and (in the mature era) internet penetration are absorbed by income, they correlate with digital-payment use unconditionally, but not separably from the income gradient. The cross-era stability itself is the meta-finding: the same pattern shows up with a five-rail composite in 2019-21 and with pure UPI in 2023-26, across different sample sizes and different time structures.

That stability points to a specific reading of the inclusion question we started with. Has UPI’s scale translated into broad-based financial inclusion? Partially yes, and the partial-yes is conditional on programmes that deliberately decouple account access from income. Where such programmes operate at scale, PMJDY is the case our data document, they leave a measurable footprint on cross-state digital-payment use that survives the income control. Where they do not, UPI’s growth is well-described as economic development, of which digital-payment activity is one expression among many. The same income gradient that explains state-level differences in education access, electricity coverage, and consumption levels also explains state-level differences in digital-payment use.

What we have established is a robust cross-state pattern. What we have not established is causation; the data and designs that would let stronger claims follow are sketched in Section 5. The cross-state record from 2019 through 2026 is consistent with a policy hypothesis worth testing. It does not yet test it.

Appendix A. Methodology and detailed results

This appendix documents the estimator, sample, and full regression results that the body of the memo summarises. Data sources and construction are in Appendix B.

A.1 Estimator and specification

The two regressions are pooled ordinary least squares with year fixed effects (early-UPI era) or year-window fixed effects (mature-UPI era) and standard errors clustered at the state level. The dependent variable and all continuous controls are log-transformed; coefficients are therefore elasticities. The fixed effects absorb the period-specific national level so the regression identifies cross-state variation rather than common time trends.

The early-UPI regression has four state-level controls: $\log(\text{per-capita NSDP})$, $\log(\text{urban share})$, $\log(\text{PMJDY beneficiaries per adult})$, and $\log(\text{bank offices per 100,000 population})$. The mature-UPI regression adds a fifth, $\log(\text{internet density per 100 population})$, that becomes available only in that period. We do not run a within-state specification: with most controls largely time-invariant in the available data, state fixed effects would absorb the variation we are trying to identify. The research design is cross-state across multiple periods, with year/window FE absorbing common time trends.

The dependent variables are definitionally different across the two regressions and are not chained. The early-era DV is per-capita transactions on a five-rail composite (BHIM + IMPS + RuPay POS + UPI + USSD); the mature-era DV is per-capita pure UPI. Each regression is interpreted within its era; the cross-era comparison in A.5 reads coefficients on the four shared controls, not the dependent variables themselves.

A note on the timing of variables within a window. The dependent variable in the mature-era regression is a 12-month average of monthly per-capita UPI within each window. The state-level controls are not 12-month averages, they are point-in-time snapshots (PMJDY beneficiaries, bank-office density, internet density), annual figures (per-capita NSDP, literacy), or interpolated values (urban share) measured at or near the relevant window. This is the standard cross-section convention in economics: outcomes are aggregated over a window, slow-moving covariates are measured at a point in time near the window. For the variables we use, the asymmetry is not a substantive concern because the controls move only modestly within a 12-month period, urban share by less than 0.5 percentage points per year, PMJDY enrollment as a slow-growing cumulative stock, bank-office density by under 5 percent per year, literacy still more slowly. The cross-state ordering of each control would be essentially the same whether we used a snapshot or a hypothetical 12-month average; for variables published only at annual or quarterly frequency, a 12-month average is not even constructible. Appendix B documents the specific snapshot date used for each control and each window.

A.2 Sample

The early-UPI regression is $35 \text{ states} \times 2 \text{ fiscal years (FY 2019-20 + FY 2020-21)} = 70 \text{ state-FY observations}$, of which 66 enter the regression after dropping two states with persistently missing per-capita NSDP data (Lakshadweep, Dadra and Nagar Haveli and Daman and Diu). The sample is 35 states rather than 36 because the source treats Jammu and Kashmir as a single unit; Ladakh did not exist as a separate union territory until October 2019 and is absent from the early-era source.

The mature-UPI regression is $36 \text{ states} \times 3 \text{ twelve-month windows} = 108 \text{ state-window observations}$. The three windows are April 2023 – March 2024, April 2024 – March 2025, and a rolling February 2025 – January 2026. The third window is rolling rather than fiscal-year-aligned because monthly data through March 2026 is not yet published; treating it as a twelve-month average preserves the same window length across observations. The regression sample drops three states with persistent missing data (Lakshadweep, Ladakh, Dadra and Nagar Haveli and Daman and Diu), leaving 99 state-window observations from 33 states.

Cluster counts are 33 in both regressions, which is at the comfortable side of the standard threshold for cluster-robust inference.

A.3 Early-UPI era regression, column-by-column build-up

The four-column build-up adds controls one at a time. Column 1 has only $\log(\text{per-capita NSDP})$ and the year fixed effects. The income elasticity is +0.95, statistically significant, and the regression already explains 54 percent of the within-year cross-state variation. Column 2 adds $\log(\text{urban share})$; the urban elasticity is +0.20 and is not significant. Column 3 adds $\log(\text{PMJDY beneficiaries per adult})$; the PMJDY elasticity is +0.53, marginally significant, and the R^2 jumps from 0.55 to 0.60, the largest single increment in the table.

Column 4, the headline, adds log(bank offices per 100,000 population); the bank-office coefficient is -0.17 , not significant. The PMJDY coefficient holds at $+0.50$.

	(1)	(2)	(3)	(4)
log(per-capita NSDP)	0.953*** (0.172)	0.821*** (0.213)	1.049*** (0.253)	1.168** (0.442)
log(urban share)		0.195 (0.168)	0.265 (0.194)	0.251 (0.218)
log(PMJ DY/adult)			0.528* (0.268)	0.501* (0.278)
log(bank offices/100k)				-0.170 (0.415)
Year FE	yes	yes	yes	yes
State-clustered SE	yes	yes	yes	yes
N	66	66	66	66
R ²	0.542	0.550	0.595	0.596

Standard errors in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4 Mature-UPI era regression, column-by-column build-up

The five-column build-up adds controls one at a time, mirroring the early-era structure. Column 1 has only log(per-capita NSDP); income elasticity is $+0.94$ with R^2 of 0.70. Column 2 adds log(internet density); the internet elasticity is $+0.27$ and is not significant. Column 3 adds log(urban share); coefficient is essentially zero. Column 4 adds log(PMJ DY/adult); the PMJDY elasticity is $+0.38$ and significant at the 5 percent level; R^2 rises from 0.70 to 0.74, again the largest single increment in the build-up. Column 5, the headline, adds log(bank offices per 100,000 population); the bank-office coefficient is -0.19 and not significant. The PMJDY coefficient holds at $+0.34$.

	(1)	(2)	(3)	(4)	(5)
log(per-capita NSDP)	0.939*** (0.113)	0.806*** (0.248)	0.806*** (0.243)	0.932*** (0.205)	1.047*** (0.311)
log(internet/100)		0.270 (0.415)	0.271 (0.454)	0.299 (0.409)	0.342 (0.415)
log(urban share)			-0.001 (0.140)	0.085 (0.111)	0.059 (0.126)
log(PMJ DY/adult)				0.383** (0.148)	0.341** (0.163)
log(bank offices/100k)					-0.194 (0.324)
Year-window FE	yes	yes	yes	yes	yes
State-clustered SE	yes	yes	yes	yes	yes
N	99	99	99	99	99
R ²	0.696	0.700	0.700	0.740	0.743

A.5 Cross-era comparison

Setting the headline columns of the two regressions side by side, on the four shared controls plus internet density (mature-era only):

Control	Early-UPI era	Mature-UPI era
log(per-capita NSDP)	1.168** (0.442)	1.047*** (0.311)
log(urban share)	0.251 (0.218)	0.059 (0.126)
log(PMJYD/adult)	0.501* (0.278)	0.341** (0.163)
log(bank offices/100k)	−0.170 (0.415)	−0.194 (0.324)
log(internet/100)		0.342 (0.415)
N	66	99
R ²	0.596	0.743

Two coefficients are positive and statistically significant in both eras: per-capita NSDP and PMJDY beneficiaries per adult. Two are statistically zero in both eras: urban share and bank-office density. None flip sign across periods. The standard-error bands on NSDP and PMJDY overlap across the two regressions, so the early-era and mature-era estimates are not statistically distinguishable from each other. Internet density appears in the mature era only and is not significant there; the asymmetric inclusion does not drive the cross-era picture.

A.6 The literacy puzzle

A sixth state-level variable, log of the literacy rate (persons aged seven and above), was tried in both regressions. We documented the result in Appendix tables A.7 and A.8 below and excluded the variable from both headline specifications on measurement-validity grounds.

What the regression does

Adding log(literacy) to the headline produces a strongly negative conditional elasticity in both eras: approximately **−4.2** in the early era (significant at 1 percent) and approximately **−2.4** in the mature era (also significant at 1 percent). Both are large magnitudes, a 10 percent relative increase in literacy would correspond to a 24 to 42 percent decrease in per-capita digital-payment use, conditional on the other controls.

The univariate relationship between literacy and per-capita digital-payment use is, by contrast, indistinguishable from zero. Regressed on its own, log(literacy) produces a coefficient of -1.57 in the early era ($p > 0.27$) and $+0.38$ in the mature era ($p > 0.77$). On their own, literacy and per-capita digital-payment use are not meaningfully correlated across Indian states.

The pattern is therefore: zero univariate relationship, large and negative conditional relationship. This is the puzzle, and it shows up in both eras.

The diagnostic

Standard explanations were checked and ruled out.

Multicollinearity. The variance-inflation factor for log(literacy) is 1.73, the lowest of any control in the full specification, well below the standard concern threshold of five. NSDP, internet density, and bank-office density all have higher VIFs. Multicollinearity is not the explanation.

Outlier states. Re-fitting the conditional regression after dropping Kerala (high literacy, mid-level UPI use), the seven small north-eastern states (high literacy, low UPI use), and both groups together moves the literacy elasticity from approximately -4.5 in the full sample to approximately -2.2 to -2.5 , still strongly negative throughout. The result is robust to obvious outlier candidates.

Sample restriction. The three states excluded for missing NSDP are too small in population to materially change a state-clustered regression. Sample restriction is not driving the result.

Functional form. Re-running with literacy in levels rather than logs gives the same qualitative pattern, large and negative conditional, near-zero univariate. The puzzle is not specific to the log specification.

A measurement-validity reading

Our reading is that the literacy variable as conventionally measured is not the variable a regression of digital-payment use should be controlling for. The Census of India and Periodic Labour Force Survey define literacy as the share of persons aged seven and above who can “read and write a simple statement with understanding.” That definition was constructed for the basic-education-access question. It is a coarse summary of cognitive engagement with text in a primary language.

A digital-payments regression would ideally control for something narrower: the share of the population who can navigate a smartphone interface, follow numeric prompts in a payments app, and complete a transaction. This depends on smartphone ownership, age structure (younger populations are more digitally fluent regardless of basic literacy), and exposure to digital workflows. The conventional literacy variable does not separate any of these from the underlying primary-language reading ability it was designed to measure.

The conditional negative coefficient is consistent with this. Once income, urban share, PMJDY enrollment, and (in the mature era) internet density are in the regression, the residual variation in literacy is what remains after the regression has absorbed the parts of literacy that do correlate with digital-payment use through those channels. What is left is whatever the variable picks up that the controls do not, and our reading is that this includes age structure, language composition, and rural-vs-urban primary-language fluency, none of which align well with smartphone-payment capability.

That the same pattern appears in both eras strengthens the diagnosis. If the literacy variable behaved differently across the two regimes, that would be evidence of an era-specific phenomenon. It does not, the same pattern, with the same direction and the same effect on PMJDY’s significance, shows up five years apart with different dependent variables. The most parsimonious explanation is that the variable is consistently measuring something other than what a digital-payments regression wants.

A better state-level control for the underlying concept, smartphone literacy, age-stratified basic literacy, or first-language literacy specifically, would let us re-run this with a variable whose conditional behaviour we could trust. That data does not currently exist at the Indian state level.

Tables

Table A.7, Early-UPI era, with literacy added

	(Univariate: literacy)	(Headline, no lit.)	(Headline + literacy)
log(per-capita NSDP)		1.168** (0.442)	0.951** (0.370)
log(urban share)		0.251 (0.218)	0.337 (0.242)
log(PMJDY/adult)		0.501* (0.278)	0.102 (0.262)
log(bank offices/100k)		−0.170 (0.415)	−0.017 (0.423)
log(literacy)	−1.574 (1.434)		−4.214*** (0.808)
N	66	66	66
R ²	0.168	0.596	0.757

Table A.8, Mature-UPI era, with literacy added

	(Univariate: literacy)	(Headline, no lit.)	(Headline + literacy)
log(per-capita NSDP)		1.047*** (0.311)	0.842** (0.314)
log(internet/100)		0.342 (0.415)	0.474 (0.378)
log(urban share)		0.059 (0.126)	0.184 (0.118)
log(PMJDY/adult)		0.341** (0.163)	0.154 (0.169)

	(Univariate: literacy)	(Headline, no lit.)	(Headline + literacy)
log(bank offices/100k)		−0.194 (0.324)	−0.063 (0.354)
log(literacy)	0.383 (1.353)		−2.383*** (0.606)
N	99	99	99
R ²	0.089	0.743	0.799

Appendix B. Data construction notes

This appendix documents the source, parsing, and construction of every variable used in Parts I and II. The full ingestion code is in `scripts/01_ingest/` and the panel-build scripts in `scripts/03_build/`.

State-name reconciliation

Every Indian data source spells state names differently. Census, RBI, NPCI, MoSPI, and Indiatat each use slightly different conventions, and within Indiatat the conventions vary by year and by report. We reconcile every incoming state name through a single canonical lookup table at `lookups/state_names.csv`. The lookup contains, for each canonical state or union territory, a pipe-separated list of every variant spelling encountered in the project. Every ingestion script joins through this table before writing to interim. Unrecognised spellings cause the script to fail loudly with the offending strings; we add new variants to the lookup and re-run rather than silently dropping rows or fuzzy-matching.

The lookup carries explicit treatment of three boundary cases. **Jammu and Kashmir** and **Ladakh** are reported as a single unit by some sources (Series 1 in Part I, RBI before 2019) and as separate units by others (post-2019 Indiatat, MoHFW). The canonical lookup keeps them as separate canonical entries; sources reporting J&K combined are aggregated to a single canonical “Jammu and Kashmir” row and Ladakh is absent from those sources. **Telangana** and **Andhra Pradesh** are separate units throughout our sample period (Telangana was bifurcated in June 2014, before the start of either Part I or Part II); no special handling is needed in our analytical period. **Dadra and Nagar Haveli** and **Daman and Diu** were merged into a single union territory in January 2020. Both pre-merger spellings appear in the lookup as variants of the merged canonical name; sources that report them as separate rows are aggregated by sum to the merged canonical.

Dependent variable, Part I (Series 1 5-rail composite)

Source: Indiatat, “State-wise Number of Digital Payment Transactions through BHIM App, IMPS, RuPay on POS, UPI and USSD”, five regional XLS files (HTML-as-XLS) covering Eastern, North-Eastern, Northern, Southern, and Western/Central India. The data originate from Lok Sabha Unstarred Question No. 1425, dated 28 July 2021. Each regional file reports an annual transaction count for FY 2019-20 and FY 2020-21 by state.

Parsing: HTML extraction via `rvest` (the files are saved as `.xls` but contain HTML tables, not Excel binary). The first table in each file contains the data; the second table contains source notes. Numeric values are read with comma stripping. Five regional files produce 36 raw rows (one per state including pre-merger Dadra and Nagar Haveli + Daman and Diu separately); after canonical reconciliation and aggregation across the merged DNHDD canonical, we have 35 unique state canonicals with two FYs each, for 70 state-FY observations.

Why FY 2017-18 is not included: an earlier Indiatat compilation (Lok Sabha Unstarred Question 5291, FY 2017-18) reports volumes on a three-rail composite, BHIM + UPI + USSD only, without IMPS or RuPay-on-POS. Chaining the two compilations would mix definitionally different DVs. We restrict Part I to the two FYs published on the consistent 5-rail composite. Decision documented in `data/raw/indiatat/series1_digital_payments/_MANIFEST.txt`.

Dependent variable, Part II (pure UPI monthly)

Source: Indiatat, “State-wise Volume and Value of Transactions Made through Unified Payments Interface (UPI) in India”, one HTML-as-XLS file per month, 35 files spanning April 2023 through January 2026. Each file reports per-state UPI transaction volume (in millions) and value (in rupees crore) for the named month.

Parsing: same `rvest` approach. Each month is a separate file; we batch-ingest all 35 into a single long table with columns `state_canonical`, `state_code`, `year_month`, `upi_volume_million`, `upi_value_crore`. The October 2024 file appears twice in the source download (one apparent duplicate); we de-duplicate at ingestion. The resulting interim file is at `data/interim/indiatat_upi_monthly.csv` with 1,260 rows (36 states $\times$ 35 months).

Per-capita normalisation in the build: for each twelve-month analytical window we compute a state’s average monthly per-capita UPI as the mean of monthly volumes within the window divided by the mid-period population (defined below).

Population

Source: Ministry of Health and Family Welfare, Population Projections for India and States 2011-2036 (the standard inter-census projection report used across Indian government statistical work). The PDF reports projected populations as of 1 July of each year from 2011 through 2036.

Parsing: `pdftools` page extraction targeting the state-wise tables. Output: `data/interim/population_mohfw.csv` with one row per state per year. Mid-period denominators in Parts I and II are computed as the mean of the two adjacent 1-July populations, placing the denominator near the midpoint of the relevant fiscal year (or rolling window).

Per-capita Net State Domestic Product

Source: Indiatat compilation of MoSPI state-account data, expressed at constant 2011-12 prices. Two ingested workbooks cover separate FY ranges with overlapping years for cross-validation; we use the values from whichever workbook reports the more recent vintage for each state-year. Output: `data/interim/per_capita_nsdp.csv` covering FYs 2011-12 through 2024-25.

Coverage by year (states reporting non-missing values): 33 for FYs 2011-12 through 2022-23 (the same three states are persistently missing, Lakshadweep, Ladakh, Dadra and Nagar Haveli and Daman and Diu); 32 for FY 2023-24 (one additional state still in advance-estimates phase); 25 for FY 2024-25 (a larger set of states delays advance-estimates publication). The build scripts apply a fallback chain, for each window’s NSDP, use the named FY first, then the prior FY, then the FY before that, to maximise observations. The three persistently-missing states are dropped from both regression samples.

Literacy

Source: Indiatat compilation of PLFS-derived state-level literacy rates (persons aged 7 and above who can read and write a simple statement with understanding). Output: `data/interim/indiatat_literacy.csv` covering FYs 2019-20 through 2023-24 plus a CY 2025 release.

Coverage: 36 states for every year in the panel. We use FY-specific values for Part I (FY 2019-20 and FY 2020-21) and FY 2023-24 carried across windows for Part II (newer literacy data is not yet available for all states on a comparable basis).

Urban population share

Source: Census 2011 enumerated urban–total population ratios as the lower anchor; Ministry of Health and Family Welfare 2021 projected urban–total population ratios as the upper anchor. The 2021 figure is a projection rather than an enumerated census because the 2021 decennial census was deferred; we use the same MoHFW report that provides the population projections.

Construction: linear interpolation between 2011 and 2021 anchors to the midpoint year of each analytical window, 2019.5 for FY 2019-20, 2020.5 for FY 2020-21, 2023.5 for Part II W1, 2024.5 for Part II W2, and 2025.5 for Part II W3. Two states (Lakshadweep and Chandigarh) interpolate to fractional values slightly above 1.0 in the post-2021 windows; we cap these at 1.0. The cap affects six state-window observations across the two parts and does not move any regression coefficient materially.

PMJDY beneficiaries per adult

Numerator: cumulative count of basic savings accounts opened under the Pradhan Mantri Jan Dhan Yojana (launched August 2014), by state, captured from snapshots of the Department of Financial Services portal page pmjdy.gov.in/statewise-statistics archived on the Internet Archive’s Wayback Machine. Five snapshots are used:

Wayback date	Used for	Distance to anchor
2019-10-17	Part I FY 2019-20	5.5 months before FY end
2021-03-05	Part I FY 2020-21	26 days before FY end
2024-02-27	Part II W1 (FY 2023-24)	33 days before window end
2025-02-15	Part II W2 (FY 2024-25)	44 days before window end
2025-12-13	Part II W3 (rolling Feb 25–Jan 26)	49 days before window end

PMJDY beneficiary counts are cumulative since 2014; cross-state rank ordering is stable across small temporal offsets, and absolute values shift modestly with calendar progression.

Denominator: adult population, derived from PLFS 2023-24 Annual Report sample composition (Table 1 of the published report; persons aged 15 and above as a share of the total survey sample, by state). The all-India adult share is approximately 0.74. PLFS 2023-24 is the only adult-share round we use; the Indian adult share moves slowly enough that applying it to all five anchors introduces small approximation error.

Bank-office density per 100,000 population

Source: Reserve Bank of India, Database on Indian Economy (DBIE), Other STRBI Table 13: “State-wise Number of Functioning Offices of Commercial Banks.” Quarterly publication; the source XLSX provides quarter-end snapshots back to March 2006. “Functioning offices” includes branches plus administrative offices and extension counters across all scheduled commercial banks (public, private, foreign, RRB, small finance, payments banks).

We extract five snapshots: 31 March 2020, 31 March 2021, 31 March 2024, 31 March 2025, and 31 December 2025. The first two go to Part I; the latter three go to Part II’s three windows (using 31 December 2025 as the closest available pre-publication snapshot for the rolling W3 window ending January 2026). Output: `data/interim/rbi_offices_state_panel.csv`.

The RBI source lists “Daman & Diu” as a separate row from “Dadra and Nagar Haveli and Daman and Diu” for some pre-merger snapshots, populating the latter with NA. Reconciliation through the canonical lookup aggregates both rows to the merged DNHDD canonical.

Internet density

Source: Telecom Regulatory Authority of India, Yearly Performance Indicators (YPI). State-level internet subscribers per 100 population, on the comparable post-2023 service-area basis. Snapshots used: 31 March 2024 and 31 March 2025. Pre-2023 TRAI data is reported by Licensed Service Areas which combine post-split states (Andhra Pradesh and Telangana, Bihar and Jharkhand, others); state-level internet density on a comparable basis is therefore unavailable for Part I.

Unclassified UPI volume, randomness test

The Indiatat NPCI state-wise files carry a “Unclassified” row in addition to the 36 state rows. The Unclassified row’s share of national UPI volume grew from approximately 12 percent in April 2023 to nearly 40 percent in January 2026. The growing Unclassified share could in principle compress observed cross-state dispersion if the loss-to-Unclassified is concentrated in particular states.

We tested this directly. For each state and month, we computed the state’s share of the national volume that is state-attributable (i.e., excluding Unclassified). If Unclassified loss were random across states, this share should be approximately stable per state across months. If it were systematically draining particular states, the share should fall fastest for those states. Pearson correlation between a state’s per-capita UPI and its rate of Unclassified-share-loss over the panel period is small ($|r| < 0.2$) and not statistically distinguishable from zero. The convergence pattern in Section 5 is robust to the bucket. Diagnostic script: `scripts/04_analyze/test_unclassified_randomness.R`.

Maharashtra share loss

One large state, Maharashtra, has lost close to six percentage points of national UPI share over the 34-month panel. This is consistent with a disproportionate share of Maharashtra’s volume migrating into the Unclassified bucket as Mumbai-based payment-infrastructure routing changes. We document this here and in Section 5 as a partial driver of the observed convergence: leaving Maharashtra out of the cross-state dispersion calculation reduces the slope of both convergence measures (CV and SD-of-log) by approximately 20 percent, leaving the qualitative downward trend intact.

Note on tools

The author drafted this paper with assistance from an AI tool (Anthropic Claude); analytical decisions and editorial direction are the author’s; all errors are the author’s own.